Noise Event Removal for Real-World Indic Speech: Team Deadlock — Track 2 (Datathon@IndoML 2026)

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***Abstract*—We describe Team Deadlock's system for Track 2 (Noise Event Removal) of Datathon@IndoML 2026, where speech degraded by non-stationary noise events must be enhanced so that SI-SDR improves and downstream ASR does not degrade. Our system is a BiGRU magnitude-mask network conditioned, via FiLM, on the frozen encoder of the mandated SraVaani-1.0 ASR model. It is trained with an SI-SDR loss on simulated mixtures and then fine-tuned on the organizers' validation (noisy, clean) pairs. On a held-out 15% slice of the validation pairs it raises SI-SDR from -0.38 dB to $+13.21$ dB, for a local Combined score of $+11.95$. We made no leaderboard submission: our final-day attempt failed with a Codabench server error after the phase had closed. All numbers are therefore local, and the system follows the organizers' published baseline design, as detailed below.**

I. Introduction

Field recordings from Indian environments contain bursty noise events (horns, animals, doorbells, appliances) that degrade ASR. Track 2 scores a submission by Combined $=$ SI-SDR$_{\text{synthetic}} + 100 \cdot \Delta$WER, where ΔWER $=$ WER(noisy) $-$ WER(enhanced) is pooled over clips and the transcripts must come from the mandated SraVaani-1.0 model run on the submitted audio. Over-suppression therefore hurts twice: it lowers SI-SDR and deletes phonemes the ASR needs.

II. Training Data and Preprocessing

Source data. Only the organizer-provided Vaani-Noise-Event-Dataset and the organizer validation set (2,501 clips, synthetic and natural) were used. No external datasets were used.

Simulated mixtures. Clean-speech spans and isolated noise-event spans are carved from annotated Gold-tier clips (streamed from Hugging Face): clean spans are regions with no annotated event, noise spans are annotated events whose speech ratio (WebRTC VAD) is below 0.35. Each mixture places one clean span (5 s windows) together with one or two noise events at an event-local SNR sampled from $[-5, 12]$ dB. We ran the smallest ("tiny") budget: 100 Gold clips scanned, yielding 43 clean and 30 noise spans, and 100 simulated mixtures (mean input SI-SDR $+9.21$ dB).

Validation pairs. The validation clips that come with a clean reference give (noisy, clean) pairs. We shuffle them with a fixed seed, hold out 15% for model selection and evaluation, and fine-tune on the rest. No augmentation beyond the simulation above was applied.

III. Proposed Method

A. Architecture

Audio is 16 kHz mono. The mask network operates on the STFT magnitude (512-point FFT, hop 128): a linear layer maps $\log(1 + |X|)$ to 256 units, followed by a 2-layer bidirectional GRU (256 hidden units, dropout 0.1), a linear+ReLU layer and a sigmoid output producing a magnitude mask per time-frequency bin. The enhanced signal is reconstructed with the mixture phase.

B. Pretrained Components

- **ARTPARK-IISc/SraVaani-1.0** (Hugging Face): its frozen FastConformer encoder (exported TorchScript graph, 128-mel front end, $8\times$ subsampling) provides conditioning features, projected to 128 dimensions and applied by FiLM (scale-and-shift) before and after the GRU. The same model is the mandated ASR used to produce `transcripts.jsonl`.
- **WebRTC VAD** (`webrtcvad`, aggressiveness 2): used only to reject noise spans dominated by speech when building the noise bank.

An ASR encoder is a deliberate choice: half the metric is ΔWER, and a conditioning signal that "knows" what speech sounds like discourages deleting it.

C. Loss and Training

The loss is the negative time-domain SI-SDR between the reconstructed waveform and the clean reference; gradients flow through the inverse STFT to the mask. Stage 1 trains on the simulated mixtures for 3 epochs (batch 8, AdamW, lr 10^{-3}, weight decay 10^{-4}, cosine schedule). Stage 2 fine-tunes on the validation pairs for 30 epochs (AdamW, lr 3×10^{-4}, cosine schedule, gradient clipping 5.0), keeping the checkpoint with the best held-out absolute SI-SDR.

D. Output Stage

$y = \alpha\, \hat{s} + (1 - \alpha)\, x$ blends the estimate with the input (α is the *blend*); the output level is restored to the input RMS (SI-SDR is scale-invariant but the ASR is not, and a sigmoid mask only attenuates); NaN/Inf are zeroed and peaks limited. Clips longer than 30 s are processed in overlapping windows with cross-fades.

E. Provenance

The architecture, simulation recipe, FiLM conditioning, loss and output stage follow the organizers' published v-1 Track 2 baseline (see references). Our work consisted of running this pipeline end-to-end on a Kaggle GPU notebook at the "tiny" budget (including its fine-tuning and blend-sweep stages), and wrapping training, inference, ASR transcription and packaging into the code released with this report. We claim no architectural novelty for Track 2.

IV. Results and Conclusion

Table I reports the blend sweep on the held-out validation slice after fine-tuning (SI-SDR on clips with a clean reference, ΔWER over clips with a reference transcript).

TABLE I
BLEND SWEEP ON HELD-OUT VALIDATION SLICE (LOCAL).

Blend α	**SI-SDR (dB)**	**ΔWER (%)**	**Combined**
0.30	+1.55	+0.09	+1.64
0.50	+3.66	+0.54	+4.20
0.70	+6.46	+1.62	+8.08
0.85	+9.21	+1.08	+10.29
1.00	**+13.21**	-1.26	**+11.95**

Stage 1 alone (simulated data only) improved SI-SDR by just $+0.89$ dB on its simulated validation split; fine-tuning on the validation pairs was decisive, raising held-out SI-SDR from -0.38 dB (noisy input) to $+13.21$ dB. The best blend is 1.0, i.e. no dry mixing: SI-SDR gains outweigh a small ΔWER loss (enhanced audio is 1.26 points worse in WER than the noisy input). Lower blends give positive ΔWER but lose more SI-SDR.

Limitations. (i) All numbers are local, on a 15% slice of the validation pairs whose other 85% was used for fine-tuning; test-set performance may differ. (ii) The tiny simulation budget leaves the stage-1 model weak. (iii) ΔWER is negative at the best blend, so ASR robustness is the main area for improvement (e.g. ASR-aware losses or a larger simulation budget). (iv) No leaderboard submission was made, so we report no rank.

References

References

[1] ARTPARK-IISc, "Vaani Corpus," https://arxiv.org/abs/2603.28714.
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[3] ARTPARK-IISc, "SraVaani-1.0," Hugging Face, https://huggingface.co/ARTPARK-IISc/SraVaani-1.0.
[4] Datathon@IndoML 2026 organizers, "Track 2 (removal) baseline v-1," https://github.com/nagshubhadip/noise-event-detection-and-removal.
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